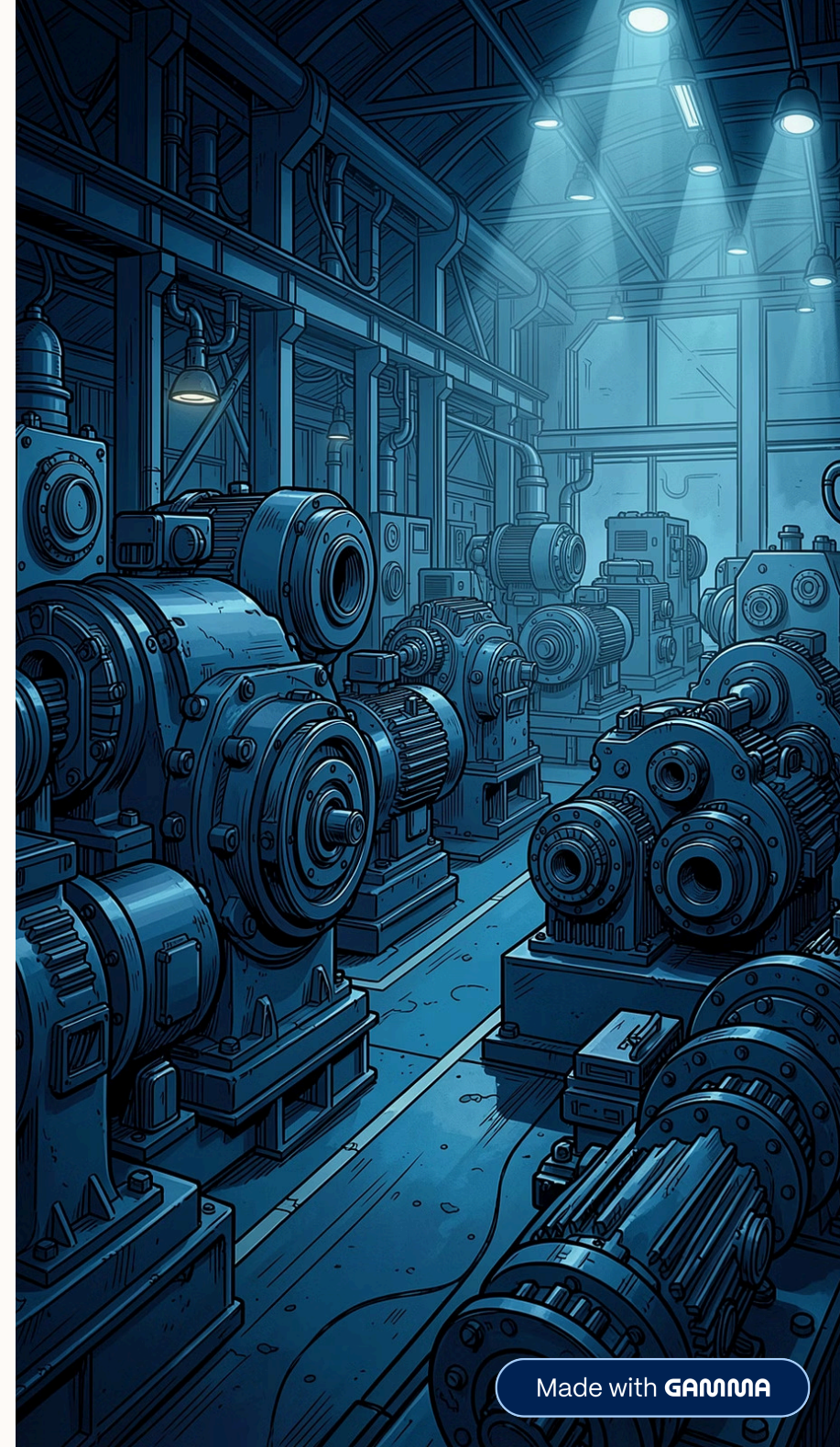


TATA TECHNOLOGIES INNOVENT-27

EdgeSense AI Real-Time Predictive Maintenance for Industrial Machinery

AI at the Edge · Industrial Heavy Machinery
By_SakinaKhan



Made with GAMMA

The Problem

Unexpected Failures

Unplanned breakdowns halt production

High Maintenance Cost

Reactive repairs drain budgets

Costly Downtime

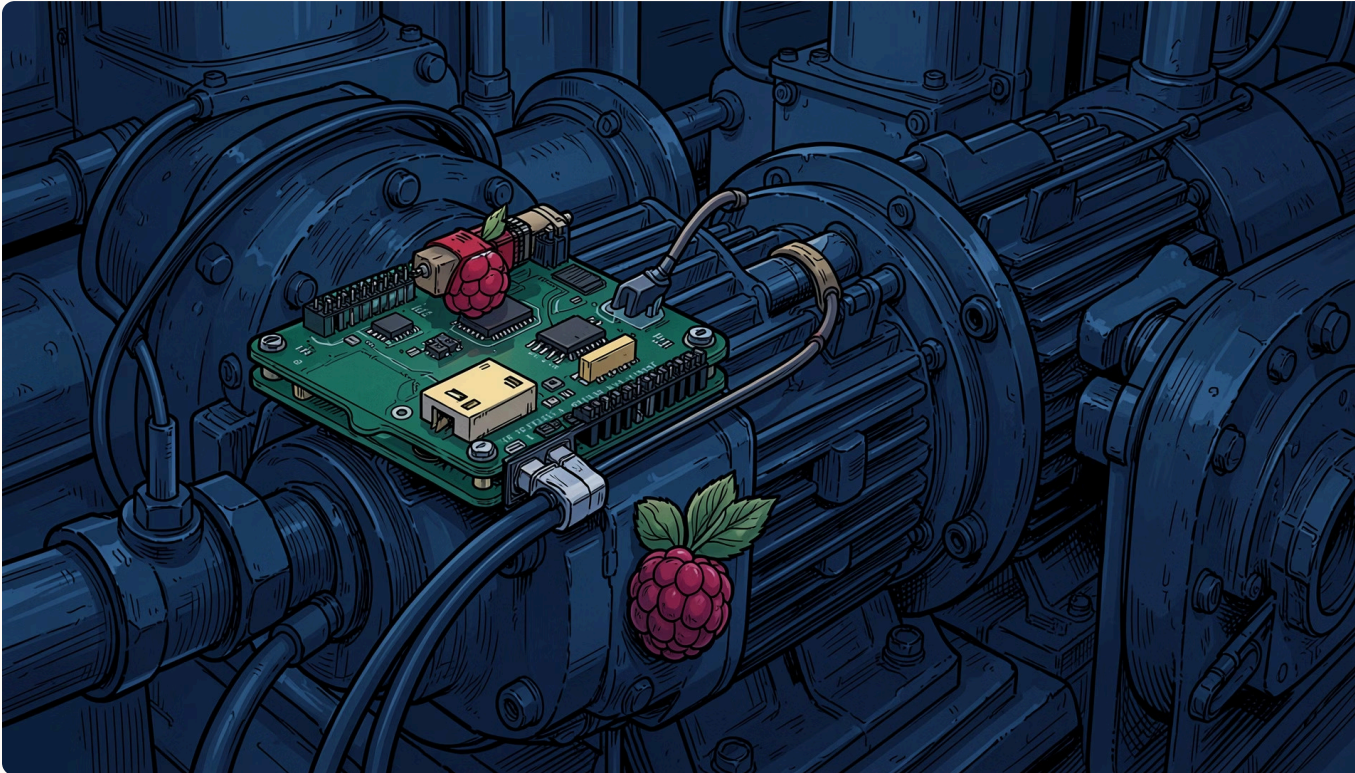
Every hour offline is lost revenue

Safety Risks

Machine failures endanger workers

Cloud Limitations

Expensive, latency-prone, internet-dependent



OUR SOLUTION

Edge AI Monitoring

EdgeSense AI continuously monitors machine **vibration and sound** using edge devices — detecting anomalies **locally, without cloud dependency**, and delivering instant alerts.

→ Sense

MEMS sensors capture vibration & audio

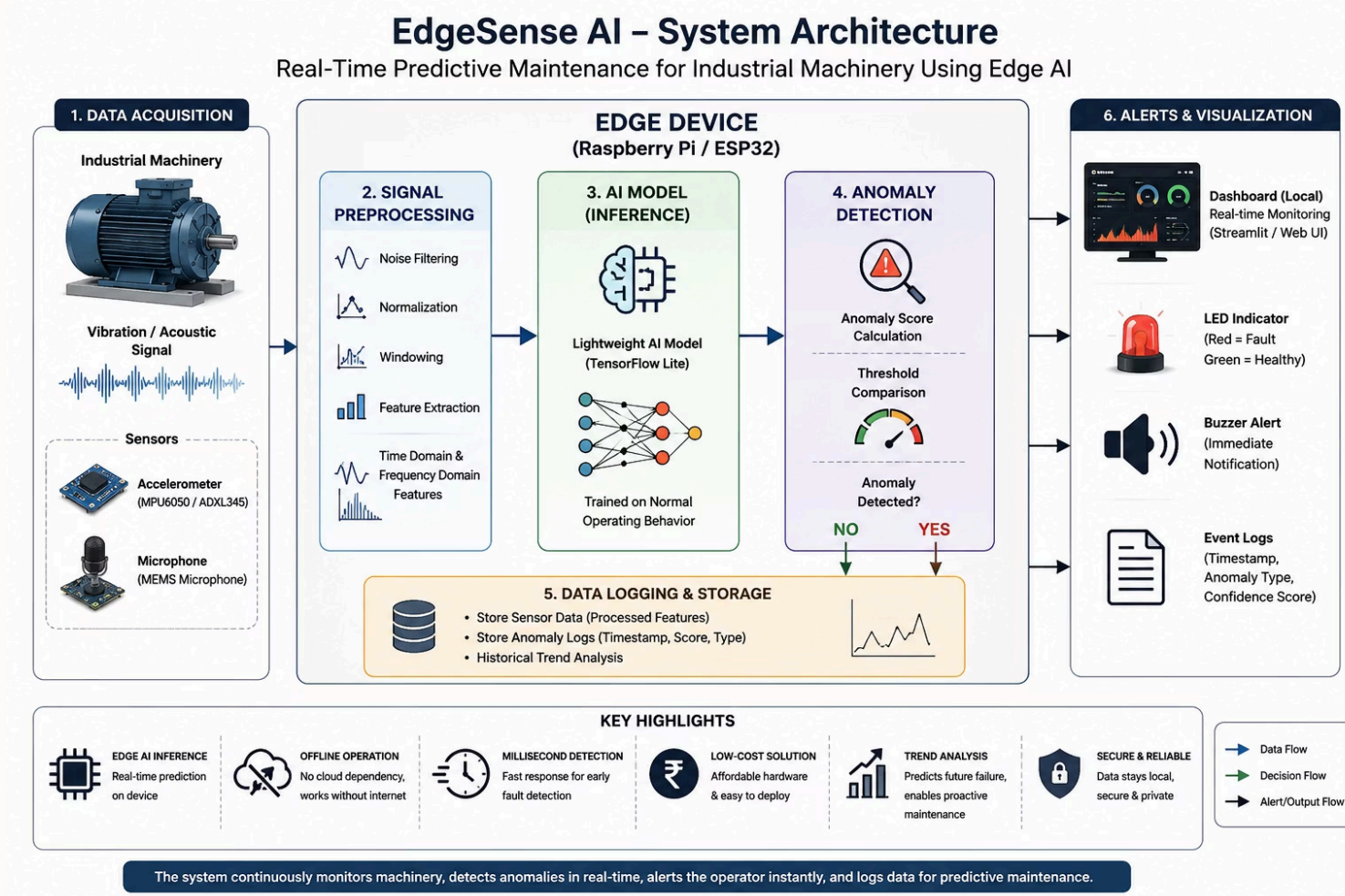
→ Process

Edge device runs TinyML inference

→ Alert

Instant dashboard notification

System Architecture



Full pipeline from raw sensor data to real-time anomaly alerts — entirely on the edge.

Technology Stack

AI / ML

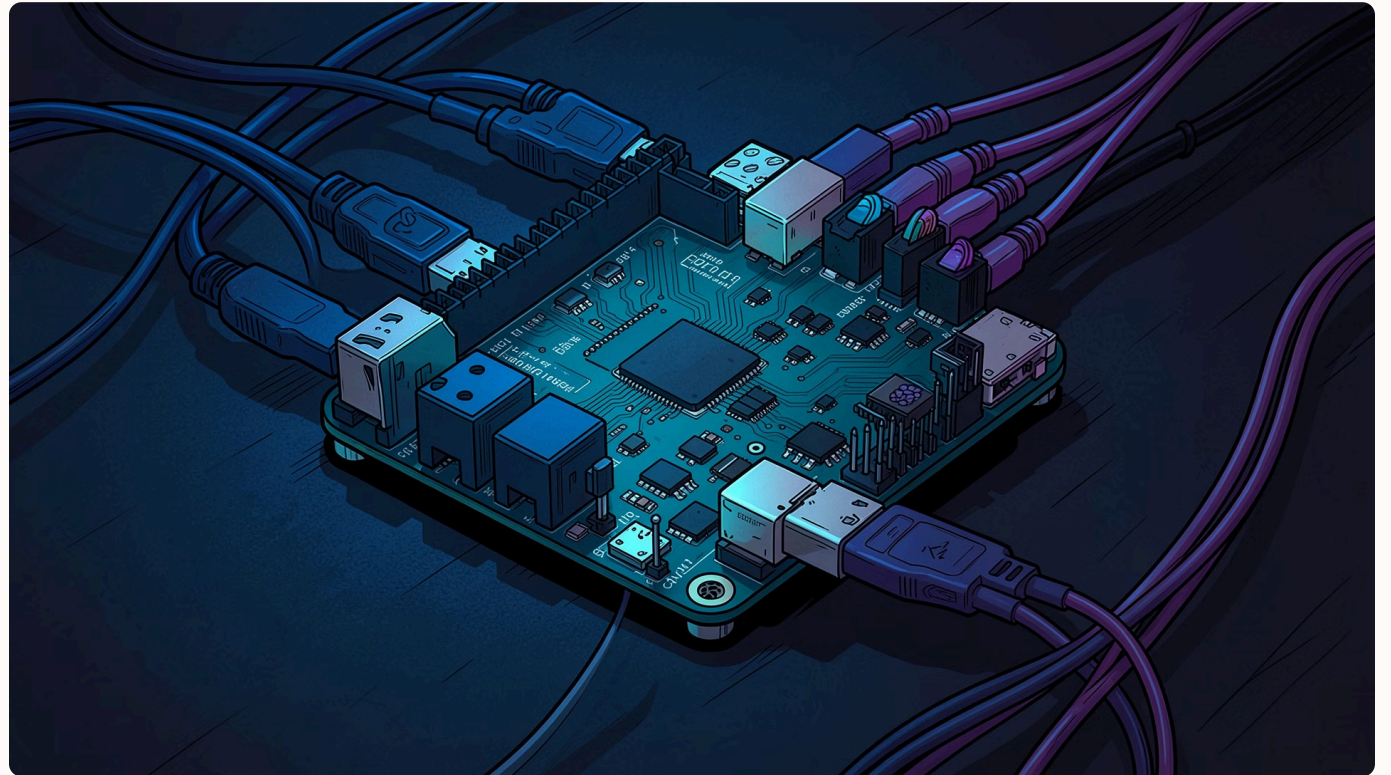
Python · TensorFlow Lite · Scikit-learn · NumPy · SciPy

Hardware

Raspberry Pi · ESP32 · MPU6050 ·
MEMS Microphone

Dashboard

Streamlit



Key Features



Edge AI · TinyML

On-device inference, no cloud



Real-Time Detection

Instant anomaly alerts



Offline Operation

Works without internet



Low Cost

Affordable for SMEs



Trend Analysis

Historical data insights



Explainable AI

Transparent decision logic

Competitive Advantage

Feature	Traditional Cloud Systems	EdgeSense AI
Cloud Dependency	Required	None
Internet Requirement	Mandatory	Not needed
Latency	High (cloud round-trip)	Ultra-low (edge)
Cost	High (subscription + infra)	Low (commodity HW)
Real-Time Processing	Limited	Yes
Data Privacy	Cloud-stored	On-premise
Scalability	Complex	Modular & easy

Progress & Roadmap

Phase 1 — Completed

Concept finalized • Architecture designed •
Technology stack selected

Phase 3 — Completed

Streamlit dashboard developed • AI model
integrated • Virtual PoC successfully
demonstrated

Phase 2 — Completed

Dataset collection • Data preprocessing • AI
model training • Model validation (99%
Accuracy)

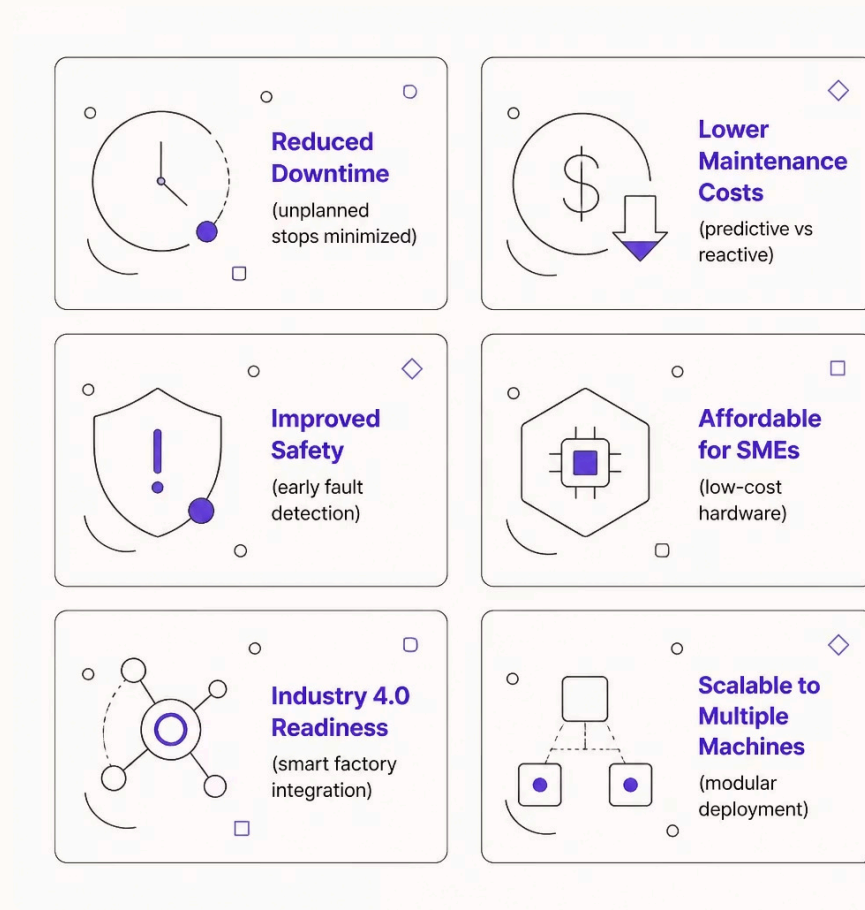
Phase 4 — Future Work

Edge device deployment (Raspberry Pi) • Live
sensor integration • Industrial field testing •
Multi-machine monitoring



Current Stage: Working Virtual PoC Completed | AI Model Integrated | Ready for Physical Prototype Development

Expected Impact



Why It Matters

- Reduced downtime & maintenance costs
- Improved worker safety
- Affordable for SMEs
- Industry 4.0 ready
- Scalable across machines

Conclusion

Solution

Edge AI monitoring on commodity hardware

Innovation

TinyML, offline, explainable, real-time

Vision

Scalable predictive maintenance for Industry 4.0

"Predict Early. Prevent Downtime. Power Smarter Industries with Edge AI."

